

PRINCIPAL AI ARCHITECTURE

AGENTIC BIOMEDICAL INTELLIGENCE & SCIENTIFIC DECISION PLATFORM

VALIDATION & REGULATORY READINESS

Principal AI Architecture — biomedical multi-agent orchestration, causal feature platform, custom training and evaluation pipelines, GraphRAG, trust and cost governance, and a human-governed GenAI operating system for mechanistic discovery, intervention prioritisation and validation.

ARCHITECTURAL POSITION

Snapshot-based · evidence-first · precompute-heavy · bounded agents
Causal provenance · explicit directionality · deterministic safety gates · accountable humans decide.

SCIENTIFIC COMPUTING · AGENTIC AI · VALIDATION CONTROL PLANE

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The first 24 pages establish the end-to-end scientific AI architecture. Pages 25–75 form a decision-engineering compendium: one explicit Principal-level packet per irreversible or safety-significant architectural choice.

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1 Problem and Architectural Position

1.1 From phenotype labels to executable mechanism hypotheses

The platform is not a biomedical chatbot and not a passive data catalogue. It is a versioned scientific decision system that converts heterogeneous evidence into falsifiable mechanism hypotheses, directionally constrained intervention candidates, validation plans and regulatory-ready evidence. The architectural unit of value is therefore not generated prose; it is a signed decision record whose evidence, transforms, gates, uncertainty and human authority can be replayed.

1.2 Why this is an AI systems problem

Failure of a conventional system	Architectural consequence
Heterogeneous identifiers, assays, contexts and release cadences	A canonical relational core, immutable lakehouse, governed graph projection and point-in-time feature compiler are all required.
Large hypothesis space and sparse labels	Candidate generation, causal fusion, learning-to-rank, calibration and value-of-information triage must work as one evaluated portfolio.
Unstructured scientific evidence and contradictory literature	Hybrid retrieval and bounded GraphRAG provide evidence; foundation models may author and challenge but never become causal or safety authority.
Direction and safety are asymmetric	Tri-state direction and deterministic safety gates execute after model review and cannot be overwritten by agents.
Validation outcomes mature slowly and contextually	Protocol, chain of custody, QC, deviation, assay context and adjudication travel with every label.

1.3 Principal architecture thesis

The governing thesis is deterministic scientific spine, agentic side branches. AWS services provide the execution substrate, but the durable architecture is the contract system: snapshot manifest, EvidenceUnit, ContextManifest, feature view, model artifact, rule bundle, hypothesis state, validation package and canonical scientific decision record.

1.4 Non-negotiable constraints

No model owns regulated or safety-critical state. No candidate is ranked before causal eligibility. No direction-dependent claim survives an UNKNOWN polarity. No safety veto is a rank feature. No partner data enters shared training by default. No retrieval result becomes evidence without source identity, admissibility and temporal applicability. No release is valid without replay, rollback and accountable human approval.

2 Reference Architecture

The architecture is deployed as an AWS multi-account scientific platform. Open and licensed source ingestion, proprietary inference, partner scoring, external validation, security evidence and regulatory packaging are separated by account and policy boundaries. Each released scientific snapshot binds the lakehouse, relational SOR, graph/search projections, feature views, model and prompt versions, rule bundles and evaluation evidence.

Plane	AWS implementation	Responsibility
Scientific data plane	Amazon S3 Object Lock, Apache Iceberg, AWS Glue, Lake Formation, Athena and EMR Serverless	Immutable source and transform history, governed discovery, point-in-time datasets and replay.
Canonical state plane	Amazon Aurora PostgreSQL, RDS Proxy, transactional outbox	Canonical entities, mappings, approvals, workflow state and decision records.
Knowledge plane	Amazon Neptune Database/Analytics and Amazon OpenSearch Service	Typed evidence graph, bounded GraphRAG, lexical/vector retrieval and contradiction mining.
Feature and model plane	SageMaker Feature Store, Pipelines, Processing, Training, managed MLflow, Registry and endpoints	Point-in-time features, custom training, calibration, batch scoring and governed model promotion.
Agentic control plane	Amazon Bedrock, Bedrock Guardrails and Bedrock AgentCore Runtime/Gateway/Identity/Observability/Evaluations	Capability routing, constrained generation, isolated sessions, policy-mediated tools and trajectory evidence.
Workflow/integration plane	AWS Step Functions, EventBridge, SQS/SNS, API Gateway and Lambda/ECS Fargate	Snapshot compiler, human waits, external validation exchange, retries, compensation and canonical APIs.
Security/operations plane	IAM Identity Center, KMS, Secrets Manager, CloudTrail, Security Lake, GuardDuty, Macie, Security Hub, Audit Manager, CloudWatch and ADOT	Zero trust, immutable audit, threat detection, scientific SLOs, incident evidence and control assurance.

2.1 Compile and serve split

Compile mode builds immutable scientific products from dated evidence. Serve mode reads released artifacts, supports bounded partner scoring and answers evidence questions. This split keeps expensive and non-deterministic work inside a governed build while preserving low-latency, rollback-safe delivery.

2.2 One decision flow

Source contracts validate releases; harmonisation resolves identifiers and context; graph and feature compilers materialise snapshot projections; causal and direction gates constrain candidates; rankers and calibrators order eligible pairs; a different Bedrock model family challenges the artifact; deterministic safety executes last; humans advance, hold or shelve; validation outcomes return as context-rich labels; the evidence packager seals the decision graph.

3 Multi-Agent Orchestration Fabric

3.1 Runtime boundary

Step Functions owns durable lifecycle. AgentCore owns bounded reasoning. The separation is semantic: anything that must survive a deployment, wait for a person or time, compensate a side effect, or provide authoritative workflow history is durable workflow; anything that assembles admissible context and proposes a typed scientific artifact is an agent task.

Agent/capability	Authority	Allowed tools	Output
Disease evidence scout	Read-only	Canonical evidence query, ontology closure, graph paths, literature retrieval	Evidence dossier with source applicability and gaps.
Endotype author	Generate hypothesis only	Context compiler and approved retrieval	Schema-valid endotype proposal; cannot see intervention tables during taxonomy authoring.
Causal verifier	Deterministic veto	Colocalisation, L2G, Mendelian and functional evidence services	PASS, FAIL or INDETERMINATE by evidence axis.
Mechanism matcher	Deterministic/model scoring	Point-in-time feature views and candidate universe	Rank-ready candidate set with feature lineage.
Direction resolver	Deterministic/abstaining	Pharmacology, signed graph and precedent	ACTIVATE, INHIBIT or UNKNOWN with conflict basis.
Bounded reviewer	Capped challenge	Evidence card and contradiction retriever	Typed defects and bounded rank adjustment; no gate writes.
Safety gate	Non-LLM and non-overridable	Label, contraindication, interaction, AE and dependency services	PASS, BLOCK or INDETERMINATE.
Evidence clerk	Deterministic	Decision graph, registries and signatures	Sealed evidence card and controlled package.
Triage copilot	Draft only	Released cards, templates and validation inventory	Advance/hold/shelve memorandum for accountable human decision.

3.2 State and termination

Agent state is typed and versioned. Every invocation carries tenant, snapshot, intended use, context manifest, allowed routes, tool capability, token/step/time/cost budgets, human takeover state and trace identifiers. State reducers make parallel merges deterministic; cycle detection and repair caps prohibit open-ended self-reflection.

3.3 Human authority

Human review is an interrupt with structured reason codes and immutable evidence, not a UI convention. Approvals, edits, vetoes and overrides become review evidence only after quality checks; no raw click is allowed to train a model automatically.

4 Biomedical Knowledge Graph and Feature Platform

4.1 Storage semantics

S3/Iceberg is replay authority; Aurora is canonical transactional state; Neptune is a governed relationship projection; OpenSearch is a retrieval projection; SageMaker Feature Store and DynamoDB provide historical and hot feature access. The same fact may appear in more than one store, but source identity and snapshot lineage remain singular.

4.2 Canonical semantics

Nodes distinguish identity, disease/endotype, intervention, evidence, assay/validation and governance artifacts. Edges distinguish association, colocalisation, genetic support, modulation, inhibition, activation, contraindication, validation and supersession. Direction, sign, evidence tier, source row/passage, licence, effective interval, confidence, context and transform lineage are mandatory where applicable.

4.3 Feature computation

Activity inference, causal fusion, tissue mediation, intervention effect, pair ranking, calibration and validation outcomes are versioned feature views. Each row carries event time, availability time, source release, transform digest, context, null policy and entity scope. Training joins select only evidence available before the decision cut-off.

4.4 GraphRAG

GraphRAG uses reviewed path templates and node/hop/time budgets. Every traversed edge hydrates to admissible source evidence. A foundation model may narrate a retrieved path; it may not invent a graph predicate, infer an unavailable edge or convert semantic similarity into causal truth.

4.5 Monitoring

Signal	Detection	Control
Source/mapping drift	Schema, row counts, identifier coverage and mapping-confidence change	Quarantine affected release; run dependency impact.
Feature drift/skew	PSI/JIS, null/default rate and online/offline parity	Freeze model promotion; recompute or demote.
Graph integrity	Illegal edge, orphan, invalid temporal interval and missing provenance	Reject graph build; retain prior graph alias.
Retrieval integrity	Recall, citation support, contradiction recall, tenant/licence precision	Block context route or corpus release.
Deletion completeness	Source-to-chunk/vector/graph/feature/model lineage traversal	Keep request open and block affected artifacts.

5 Model Portfolio and Training

5.1 Portfolio

Capability	Model/control	Release evidence
Causal evidence fusion	Calibrated stacker over independent evidence axes	Temporal/group splits, correlated-source diagnostics, calibration and sentinel recovery.
Biomedical representations	Domain encoders and graph embeddings	Retrieval/cluster utility, neighbourhood stability, domain and subgroup robustness.
Taxonomy	Leiden/HDBSCAN or constrained clustering plus stability tests	Bootstrap stability, coherence, coverage and explicit outlier policy.
Direction	Signed rules/network propagation with abstention	Zero wrong-direction on sentinel set; conflict and UNKNOWN behaviour.
Ranking	Monotone GBDT or controlled ranker	NDCG/enrichment, novelty splits, monotonicity, stability and prospective yield.
Uncertainty	Isotonic/Platt and conformal methods	ECE/Brier, empirical coverage and subgroup undercoverage.
Safety	Deterministic rule engine plus narrow classifiers	Gate recall, false-block analysis, mutation tests and dependency outage.
Generative	Bedrock author/reviewer/narrator routes	Schema, grounding, contradiction handling, injection resilience and incremental value.

5.2 Training pipeline

SageMaker Pipelines compiles a rights-approved point-in-time dataset, validates leakage boundaries, builds grouped temporal splits, trains and tunes, calibrates, evaluates robustness and subgroup behaviour, scans the container/model package, registers immutable artifacts and creates data/model cards. Promotion is a separate workflow requiring scientific, model-risk, security, quality and operational evidence.

5.3 Evaluation hierarchy

Evaluation spans data contracts, feature fixtures, custom-model metrics, generated artifact validity, agent trajectory correctness, complete scientific-system sentinels, validation-loop outcomes and human factors. Aggregate accuracy is never an adequate release decision. Hard gates—direction, safety, tenant isolation, licence and lineage—have zero-tolerance acceptance criteria.

5.4 Leakage firewall

Leakage controls apply at code, data, entity lineage, time, retrieval, foundation-model cut-off and publication. Evaluation uses opaque identifiers and masked versus unmasked cohorts where parametric memory could know showcase interventions. Any model route whose training cut-off cannot support the claim is excluded from that benchmark.

6 AI Control Plane and Retrieval

6.1 Gateway and router

Every Bedrock and SageMaker invocation passes one gateway that resolves capability to evaluated route, compiles minimum admissible context, enforces structured output, records cost and trace, applies guardrails, validates semantics and chooses a tested fallback. Agents name capabilities—not model IDs.

6.2 Context compiler

The compiler emits a signed ContextManifest containing exact evidence references, versions, policy predicates, data classes, licence obligations, token allocation, omissions and context hash. Authoritative instructions, structured facts, retrieved passages and untrusted text remain separate labelled channels.

6.3 Retrieval strategy

Exact structured facts, lexical/vector passages and bounded graph paths are retrieved in parallel under pre-compiled access policy. Fusion and reranking retain raw channel scores. Context allocation reserves space for contradiction, negative evidence and source diversity, because scientific retrieval must challenge as well as support.

6.4 Guardrail stack

Layer	Enforcement
Authority	Agent manifests prohibit writes to causal, direction, safety, approval and released-snapshot state.
Syntax/semantics	Constrained output plus JSON Schema, ontology, identifier, unit, range and cross-field checks.
Grounding	Field-level source support, source existence, entailment and contradiction status.
Security	Instruction/data separation, tool allow-list, capability token, egress/DLP and injection canaries.
Regulatory	Intended-use/claim library, prohibited-use policy, controlled-template and approval checks.
Economics	Preflight estimate, token/step/retrieval/graph/time budgets and decision-level cost ledger.

7 LLMOps, MLOps and Scientific AI SRE

7.1 Artifact operations

Models, datasets, feature definitions, graph builds, prompts, schemas, agent manifests, model routes, retrieval corpora, rule bundles and validation sets are first-class artifacts with independent owner, intended use, lineage, evaluation, compatibility and rollback. A released snapshot resolves all aliases once and records the immutable versions.

7.2 Scientific SLOs

SLO	Control intent
Decision trace completeness	Every surfaced candidate links to snapshot, evidence, features, models, prompts, rules, safety and human disposition.
Hard-gate integrity	Zero gate bypass in release, replay and adversarial suites.
Direction sentinel	Zero wrong-direction promotion; UNKNOWN preserved.
Grounded generation	Task-specific citation/field support and contradiction handling above approved floor.
Snapshot reproducibility	Deterministic stages bit-reproducible; generated artifacts replayable from cache/artifact tuple.
Validation lineage	Every label links to protocol, run, raw data, QC, deviation and adjudication.
Recovery readiness	Prior signed snapshot and deterministic serving remain available during AI/control-plane failure.

7.3 Incidents

Incident severity follows scientific harm, data rights and evidence integrity—not only downtime. Gate bypass, fabricated provenance, cross-tenant disclosure, systematic unsupported claims, wrong direction, corrupt validation results and unrecoverable snapshot state each have dedicated kill switch, evidence-preservation and impact-analysis runbooks.

7.4 Progressive delivery

A change moves through unit/contract, frozen replay, shadow snapshot, scientific canary, released snapshot and independent rollback. Model, prompt, graph, feature, rule and retrieval aliases can be reverted independently, but an interaction defect reverts the complete candidate snapshot.

8 Security, Compliance, Resilience and Cost

8.1 Zero trust and data protection

Federated human identity, workload IAM roles, short-lived credentials, private service endpoints, controlled egress, purpose/tenant claims, KMS encryption context, row/partition/index isolation and immutable audit are baseline controls. Agents receive no database credentials. Confidential identifiers are placeholder-substituted before any route not approved for clear-text use.

8.2 Threat model

Primary threats include prompt and retrieval injection, evidence/graph poisoning, falsified citations, model/label leakage, partner re-identification, safety-rule tampering, forged validation results, denial of wallet and unauthorised regulatory claims. Controls are independently testable and sit outside the model wherever authority or data movement is involved.

8.3 Resilience

Capabilities are tiered. Safety rules, signatures, kill switches and audit intake remain highly available and fail closed. Released snapshot and evidence-card reads recover before interactive GraphRAG, which recovers before compilation and training. A multi-Region continuity workflow measures actual RTO/RPO through game days and preserves deterministic serving when model routes are unavailable.

8.4 Cost governance

Cost is attributed to snapshot, tenant, capability, agent, hypothesis and decision. Budgets control feature recomputation, training experiments, Bedrock tokens, retrieval depth, graph expansion, validation capacity and retention. Optimisation may narrow optional analysis or route to smaller evaluated models; it may never weaken safety, access policy or evidence integrity.

9 Canonical APIs and Decision Record

Canonical APIs prevent agents, partner services and validation integrations from inventing incompatible scientific objects. Every mutation requires identity, tenant, intended use, idempotency key, expected artifact version, snapshot and trace context.

Contract	Purpose	Invariant
EvidenceUnit	One admissible structured fact, passage or graph path	Immutable source, applicability, policy and support status.
EndotypeProposal	Structured mechanism hypothesis	Drug-blind authoring boundary and missing-evidence declaration.
CandidateScore	Eligibility, direction, raw rank, calibration and safety references	No surfaced candidate without exact gate artifacts.
ValidationPackage	Hypothesis, protocol, materials, assays, endpoints, acceptance and approvals	Package hash binds work transferred externally.
ValidationResult	Raw references, QC, deviations, effects, uncertainty and signer	No training label before maturity/adjudication.
RegulatoryEvidencePackage	Released snapshot, decisions, validation and controlled artifacts	Deterministic traceability matrix; generated prose cannot replace records.
ScientificDecisionRecord	Append-only end-to-end decision graph	Corrections supersede; they never overwrite history.

9.1 Error semantics

Errors such as EVIDENCE_INSUFFICIENT, DIRECTION_UNKNOWN, SAFETY_BLOCKED, SNAPSHOT_MISMATCH, LICENCE_DENIED, STALE_FEATURE, VALIDATION_QUARANTINED and POLICY_DENIED are domain states with permitted recovery. They are not collapsed into generic HTTP failure or hallucinated away by an agent.

10 Architecture Governance and Delivery

10.1 Decision method

Architecture starts from the decision inventory, not a service inventory. Every decision is classified by reversibility, scientific authority, data rights, latency, failure asymmetry and accountable human. Hard scenarios are traced before technology selection, contested choices are spike-tested against pre-agreed metrics, and the outcome event/label is designed with the production decision.

10.2 Delivery sequence

Phase	Scope	Exit evidence
0 Control foundations	Landing zone, identity, data classes, audit, schemas and source/licence registry	Threat model, isolation tests and immutable audit pass.
1 Snapshot evidence core	S3/Iceberg, harmonisation, Aurora, manifest and released read model	Rebuild and atomic rollback demonstrated.
2 Graph and feature compiler	Neptune/OpenSearch projections, PIT feature views and dataset compiler	Golden fixtures, skew and leakage tests pass.
3 Deterministic scientific models	Causal/direction/safety gates, ranker, calibration and uncertainty	Sentinel, subgroup, calibration and hard-gate suites pass.
4 Governed GenAI	Bedrock gateway, context compiler, AgentCore tools, author/reviewer	Grounding, authority, injection, budget and trajectory suites pass.
5 Human scientific operating model	Review queues, evidence cards, approvals, change control and kill switches	Reviewer usability, SLA and audit completeness proven.
6 Validation closed loop	Protocol packages, CRO exchange, result/QC/deviation/adjudication	Chain of custody and prospective validation run pass.
7 Regulatory readiness	Traceability, model/data cards, validation reports and controlled dossier	Mock audit and evidence-package review pass.

10.3 Principal acceptance criterion

Every scientific conclusion can be traced to admissible evidence, reconstructed from a released snapshot, challenged by independent controls, bounded by uncertainty and safety, approved by accountable humans, and carried into validation or regulatory use without trusting a model's narrative about its own reasoning.

11 Decision Engineering Compendium

Each following page is an explicit architecture decision packet. The repetition of WHY, WHERE and WHEN is intentional: these fields convert design opinion into enforceable operating architecture and make review, implementation, incident response and future reconsideration unambiguous.

Jyotirmoy Bardhan

DE-01 Immutable scientific snapshot compiler

Dimension	Principal-architecture position
Decision	Compile source releases, mappings, feature views, graph projections, models, prompts, rules and safety bundles into one signed release unit.
WHY	Biomedical evidence changes asynchronously; releasing components independently creates combinations that were never evaluated together. Snapshot coupling converts a moving knowledge system into a testable scientific product.
WHERE	The boundary spans S3/Iceberg inputs, Glue catalog versions, Aurora canonical facts, Neptune/OpenSearch projections, SageMaker artifacts, Bedrock routes and the released evidence mart.
WHEN	Run for scheduled source refresh, critical safety update, approved model/rule change or partner-specific evidence package; never mutate an already released snapshot.
Execution mechanics	A Step Functions compiler reads a manifest, invokes idempotent build stages, persists content-addressed outputs, runs cross-artifact evaluation and atomically moves a serving alias only after approvals.
Rejected alternative	Independent live updates and in-place database migration were rejected because rollback could not restore the exact scientific state consumed by a prior decision.
Failure / degraded mode	Partial build, mismatched alias or missing source release keeps the candidate snapshot unreleased; serving remains on the previous signed alias.
Validation evidence	Rebuild from the manifest, compare hashes, replay sentinel hypotheses and prove alias rollback without recomputation.
Revisit trigger	Revisit when refresh latency, storage multiplication or emergency-safety propagation violates an approved service objective.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

DE-02 AWS multi-account scientific landing zone

Dimension	Principal-architecture position
Decision	Separate ingestion, open evidence, proprietary intelligence, partner workloads, validation operations, security tooling and regulated evidence into policy-isolated AWS accounts.
WHY	Account boundaries reduce blast radius, make licence and tenant policy enforceable outside application code, and let security evidence reflect actual administrative separation.
WHERE	AWS Organizations and Control Tower define organisational units; SCPs, IAM Identity Center, KMS key hierarchy, private networking and central CloudTrail/Security Lake span the estate.
WHEN	Establish before source onboarding or partner data ingestion; create a dedicated account when legal rights, residency, operational ownership or recovery objectives diverge.
Execution mechanics	Provision through CDK/CloudFormation with mandatory tags, private endpoints, central DNS, egress controls, backup policy and delegated security administration.
Rejected alternative	A single account with namespace separation was rejected because privileged operators and broad service roles could cross data-right and partner boundaries.
Failure / degraded mode	If policy deployment or central identity fails, workload accounts continue serving released snapshots but promotion, new access grants and cross-account transfers stop.
Validation evidence	Continuously evaluate SCP conformance, route-table reachability, public-resource exposure, KMS grants and cross-account role assumptions.
Revisit trigger	Revisit cell/account granularity when tenant count, partner contractual isolation or operational overhead materially changes.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-03 S3 and Iceberg as replay authority

Dimension	Principal-architecture position
Decision	Use versioned Amazon S3 objects and Apache Iceberg tables as the immutable replay authority; databases, indexes, graphs and feature stores remain rebuildable projections.
WHY	Scientific reproducibility requires source-native payloads, event-time history and non-destructive correction. Operational stores optimise access but cannot be the only copy of transformed evidence.
WHERE	Raw, harmonised, feature, training, evaluation, validation and regulatory zones use separate buckets, KMS contexts, Lake Formation permissions and lifecycle policies.
WHEN	Write once at ingestion and after every deterministic transformation; publish a table snapshot only after schema, quality, licence and lineage checks pass.
Execution mechanics	Object keys are content addressed; Iceberg snapshots reference exact files; Glue/Lake Formation govern discovery; snapshot manifests retain table snapshot IDs and executable digests.
Rejected alternative	Database backups alone were rejected because they do not express transformation lineage, point-in-time joins or selective reconstruction of derived products.
Failure / degraded mode	Corrupt projections are deleted and rebuilt from the last trusted Iceberg snapshot; corrupt source objects are quarantined without rewriting prior history.
Validation evidence	Perform restore drills, manifest-to-object completeness checks, time-travel queries and row-level reconciliation against generated projections.
Revisit trigger	Revisit file partitioning and compaction when scan amplification, metadata growth or snapshot-retention cost exceeds the operating envelope.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-04 Aurora PostgreSQL canonical system of record

Dimension	Principal-architecture position
Decision	Use Aurora PostgreSQL for canonical identities, mapping decisions, workflow state, approvals, compact evidence facts and referential constraints.
WHY	The platform needs transactions, uniqueness, temporal validity, foreign keys and accountable state transitions. Graph and search engines do not provide the same integrity semantics.
WHERE	Aurora owns canonical entity keys, bridge tables, source-release registration, decision records, human review, protocol state and artifact aliases; bulk omics remains in the lakehouse.
WHEN	Write during harmonisation, review, validation and release transitions; analytical scans and model training read exported or lakehouse projections.
Execution mechanics	RDS Proxy bounds connections; row security and tenant predicates protect partner records; outbox rows publish domain events only after the committing transaction succeeds.
Rejected alternative	DynamoDB-only and graph-as-system-of-record designs were rejected because multi-entity constraints, temporal joins and controlled approvals became application-managed.
Failure / degraded mode	Fail over within the cluster; if writes are unavailable, freeze state-changing workflows and keep read-only snapshot serving active.
Validation evidence	Test transactional invariants, outbox exactly-once effects, tenant RLS, PIT restore and schema migration rollback against production-scale copies.
Revisit trigger	Revisit when a bounded context demonstrates independent scale and consistency semantics that justify extraction from the relational core.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-05 Neptune as governed evidence-graph projection

Dimension	Principal-architecture position
Decision	Materialise semantically valuable causal, mechanistic, intervention, safety and validation relationships in Amazon Neptune while retaining source truth in the lake and relational SOR.
WHY	Typed multi-hop questions are expensive and opaque as repeated joins. A graph projection improves traversal and explanation without conflating derived edges with authoritative source records.
WHERE	Neptune Database serves bounded operational traversals; Neptune Analytics runs centrality, community, path and vector-graph analysis against a frozen graph snapshot.
WHEN	Rebuild per scientific snapshot; use request-time traversal only for approved path shapes whose latency and access behaviour are bounded.
Execution mechanics	The graph compiler validates node/edge schema, temporal validity, provenance and tenant predicates; every derived path hydrates back to source rows or passages.
Rejected alternative	A universal graph storing every observation was rejected because write amplification, licence filtering and semantic ambiguity would dominate operational value.
Failure / degraded mode	On graph outage, exact structured retrieval and precomputed explanation subgraphs remain; no gate may infer missing graph evidence.
Validation evidence	Validate edge-count reconciliation, illegal-edge detection, orphan rate, path gold sets, tenant isolation and graph-to-source hydration.
Revisit trigger	Revisit edge materialisation when query telemetry shows stable new relationship patterns or projection cost exceeds its explanation value.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-06 OpenSearch hybrid biomedical retrieval

Dimension	Principal-architecture position
Decision	Use Amazon OpenSearch for rights-aware BM25, vector recall, filters and reranking over literature, labels, protocols and released evidence artifacts.
WHY	Biomedical retrieval needs exact identifiers and terminology as well as semantic similarity. Neither vector-only nor lexical-only search provides reliable recall across nomenclature and long-tail scientific language.
WHERE	Indexes are partitioned by corpus, publication state and security domain; tenant-specific validation and partner content never share a global semantic cache.
WHEN	Build after corpus approval and embedding evaluation; query only after identity, intended-use, licence, effective-date and snapshot filters are compiled.
Execution mechanics	Retrieve lexical and vector candidates in parallel, fuse scores, rerank with a biomedical cross-encoder and return immutable EvidenceUnits with passage and applicability metadata.
Rejected alternative	Vector search as the evidence authority was rejected because proximity does not establish entailment, causality, temporal relevance or licence permission.
Failure / degraded mode	Degrade to filtered lexical retrieval; generative claims requiring unavailable semantic evidence are withheld or routed to human review.
Validation evidence	Measure evidence recall, nDCG, citation entailment, source diversity, contradiction recall, tenant precision and deletion completeness.
Revisit trigger	Revisit embedding, fusion and sharding when drift or corpus growth changes quality, latency or operational cost.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-07 Point-in-time feature platform

Dimension	Principal-architecture position
Decision	Use SageMaker Feature Store with S3 offline history and DynamoDB hot projections for declared, versioned, point-in-time biomedical feature views.
WHY	Training-serving skew and temporal leakage invalidate scientific performance claims. Feature definitions need ownership, event time, availability time, entity scope and lineage.
WHERE	Offline views support dataset compilation and batch scoring; online projections serve partner compounds and interactive decisions; massive matrices stay in Iceberg.
WHEN	Compute after harmonisation and graph publication; retrieve at the decision as-of time, not latest value, for training, replay and audit.
Execution mechanics	Feature definitions are code-reviewed packages with transform digest, source snapshot, null policy, context dimensions and licence class; registry promotion runs golden fixtures.
Rejected alternative	Ad hoc SQL features and latest-value joins were rejected because results could not be reproduced and future evidence leaked into historic decisions.
Failure / degraded mode	Stale or missing critical features cause explicit STALE_FEATURE or EVIDENCE_INSUFFICIENT states; prohibited defaults are never silently injected.
Validation evidence	Run point-in-time join tests, sampled online/offline parity, distribution drift, lineage traversal and entity-scope leakage tests.
Revisit trigger	Revisit online-store placement when access patterns, freshness or per-tenant isolation cannot be met economically.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-08 Durable workflow separated from agent reasoning

Dimension	Principal-architecture position
Decision	Use Step Functions for durable scientific lifecycle and Bedrock AgentCore for bounded reasoning inside a workflow task.
WHY	Human waits, retries, CRO transfer and compensations have different semantics from model reasoning. Combining them creates non-deterministic recovery and duplicated side effects.
WHERE	Step Functions owns snapshot state, approvals, timers, validation exchange and release; AgentCore owns short-lived evidence assembly, authoring, critique and tool trajectories.
WHEN	Use a durable workflow whenever work must survive deployment, wait for time or a person, or compensate. Use an agent only to produce a proposal from bounded context.
Execution mechanics	Each agent invocation is an idempotent workflow task with input/output schema, deadline, heartbeat, trace, budget and callback token for approved interrupts.
Rejected alternative	A monolithic agent loop was rejected because it could not guarantee termination, replay or exactly-once external effects.
Failure / degraded mode	Agent failure returns a typed task result; workflow retries an alternate route or degrades. Workflow history remains authoritative.
Validation evidence	Replay workflow history, inject timeout/duplicate callbacks, verify idempotency, and score trajectory validity independently from business-state correctness.
Revisit trigger	Revisit runtime boundaries only when a responsibility changes durability or side-effect semantics.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-09 AgentCore identity and tool mediation

Dimension	Principal-architecture position
Decision	Run agents with workload identity and expose scientific capabilities through AgentCore Gateway/MCP-compatible tools rather than direct data credentials.
WHY	A prompted model is not a security principal. Tool mediation constrains verbs, entities, parameters, budgets and data classes independently of model intent.
WHERE	Every agent tool path—from evidence query and graph expansion to safety check and validation package—passes through the gateway and policy decision point.
WHEN	Evaluate policy before tool discovery and again before invocation; issue short-lived capability tokens for one workflow, tenant, tool and purpose.
Execution mechanics	Tool contracts use JSON Schema, deterministic validators, parameterised queries, result-size limits and immutable audit spans; agents receive references instead of broad payloads.
Rejected alternative	Direct database access and vendor-native unconstrained function calling were rejected because least privilege and portable evaluation were not enforceable.
Failure / degraded mode	Denied or unavailable tools produce typed errors; agents cannot search for alternate side channels. Critical missing tools terminate or escalate the workflow.
Validation evidence	Red-team parameter smuggling, confused deputy, token replay, tenant substitution, graph explosion and result-based prompt injection.
Revisit trigger	Revisit tool granularity when policies become either too broad to isolate risk or too fine-grained to operate reliably.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

DE-10 MCP internally and signed A2A at the boundary

Dimension	Principal-architecture position
Decision	Use MCP-compatible typed tools inside the trust boundary and signed A2A-style task/evidence envelopes only for CRO or partner exchange.
WHY	Internal free-form agent conversation adds no scientific authority and weakens replay. Organisational boundaries require explicit delegation, acknowledgement and evidence transfer semantics.
WHERE	MCP terminates at the internal tool gateway. External exchange terminates at API Gateway/EventBridge integration accounts with mTLS, signature and tenant policy.
WHEN	Use MCP for synchronous read/tool capability; use A2A only when another organisation owns execution and returns a separately accountable artifact.
Execution mechanics	External envelopes include task ID, intended use, input manifest, schema, classification, expiry, callback, signer and evidence hash. No transitive internal capability is delegated.
Rejected alternative	A2A for all internal communication was rejected because it replaces typed in-process state with distributed conversational trust.
Failure / degraded mode	Unverified, expired or schema-incompatible external responses are quarantined; workflow waits or follows an approved manual route.
Validation evidence	Contract-test signatures, replay protection, acknowledgement, duplicate delivery, schema evolution, non-repudiation and cross-tenant routing.
Revisit trigger	Revisit the boundary when a partner integration requires delegated tool use rather than artifact exchange.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-11 Scientific context compiler

Dimension	Principal-architecture position
Decision	Build the minimum admissible context for every model invocation from snapshot-bound EvidenceUnits rather than allowing agents to browse unconstrained stores.
WHY	Context selection controls hallucination, leakage, cost and reproducibility. Prompt instructions cannot compensate for excessive or unauthorised evidence.
WHERE	The compiler sits between the model gateway and Aurora, Neptune, OpenSearch and feature services; it emits a signed ContextManifest.
WHEN	Compile after capability routing and before every model call; recompile when tenant, intended use, snapshot, model route or policy changes.
Execution mechanics	Apply source, licence, date, tenant and drug-blind filters; allocate tokens by evidence tier; label instructions versus untrusted text; record hashes and omissions.
Rejected alternative	Direct retrieval inside the model and whole-document prompt stuffing were rejected because access, cut-off and citation semantics became invisible.
Failure / degraded mode	If critical evidence cannot fit or be retrieved, return context incomplete and prevent a complete-answer claim; deterministic templates remain available.
Validation evidence	Replay context hashes, test minimum-necessary inclusion, injection separation, omitted-evidence disclosure and token-budget determinism.
Revisit trigger	Revisit compression and allocation when model context capacity changes, but retain the manifest and access invariants.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-12 Capability-based Bedrock model routing

Dimension	Principal-architecture position
Decision	Route requests by evaluated capability, risk, data class, latency and cost rather than hard-coded model name.
WHY	Foundation-model quality and availability change faster than scientific workflows. Capability indirection enables controlled substitution without rewriting agent logic.
WHERE	The Bedrock gateway resolves author, reviewer, extractor and narrator classes to approved inference profiles and Regions; no LLM route exists for gates.
WHEN	Select before context compilation; fail over only to a route evaluated for the same output schema, policy class and data classification.
Execution mechanics	Router policy stores quality floor, max latency/cost, model-family diversity, fallback order and zero-retention/private-connectivity requirements.
Rejected alternative	Single-model binding was rejected because outage, model retirement and correlated author-judge errors became platform incidents.
Failure / degraded mode	If no route satisfies policy, omit generative promotion and preserve deterministic ranking or human work queues.
Validation evidence	Replay frozen traces across candidate models; evaluate schema, support, contradictions, latency, cost and route consistency.
Revisit trigger	Revisit on model version change, material drift, price change, capacity failure or new data-governance constraint.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-13 Independent author and reviewer model families

Dimension	Principal-architecture position
Decision	Use a different evaluated model family for bounded scientific review than for hypothesis authoring.
WHY	A generator judging its own output exhibits correlated blind spots; diversity is a low-cost control for unsupported claims and reasoning shortcuts.
WHERE	Applied to endotype authoring, rescue hypotheses, evidence narratives and regulatory drafts; deterministic validators run before and after review.
WHEN	Invoke only for artifacts crossing a release or human-review threshold; low-risk extraction uses deterministic checks without unnecessary second-model cost.
Execution mechanics	Reviewer receives the artifact, exact evidence set, contradiction set and defect rubric—not the author hidden reasoning—and emits typed defects and capped adjustments.
Rejected alternative	Self-critique alone and unrestricted multi-model voting were rejected because they either correlate errors or obscure accountability.
Failure / degraded mode	Reviewer outage means no generative promotion; base deterministic output remains. Reviewer disagreement increases uncertainty or human review.
Validation evidence	Measure incremental defect recall, false veto, correlated-error rate, rank movement and cost per accepted correction.
Revisit trigger	Revisit model pairing when evaluation shows convergence, systematic bias or inadequate biomedical terminology performance.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-14 Schema-constrained scientific generation

Dimension	Principal-architecture position
Decision	Require generated endotypes, reviews, protocols and dossier sections to conform to versioned schemas with semantic validators.
WHY	Free text cannot safely drive downstream computation. Typed output separates machine state from narrative and makes prohibited field writes observable.
WHERE	Enforced at Bedrock invocation, AgentCore state reducers, canonical APIs and controlled-document assembly.
WHEN	Use for every machine-consumed output; narrative is generated only from an already validated structured artifact.
Execution mechanics	Constrained decoding is followed by identifier, ontology, unit, range, source-existence, authority and cross-field validation. One repair is allowed before quarantine.
Rejected alternative	Regex parsing and best-effort JSON were rejected because malformed or semantically invalid fields became latent scientific state.
Failure / degraded mode	Invalid objects are not partially committed. Quarantine retains raw output and defects for evaluation without contaminating training labels.
Validation evidence	Track syntactic validity, semantic validity, repair success, prohibited-write attempts and schema-version compatibility.
Revisit trigger	Revisit schema granularity when fields cannot express uncertainty or when optionality is being used to hide missing critical evidence.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

DE-15 Deterministic causal eligibility gate

Dimension	Principal-architecture position
Decision	Require explicit causal evidence disposition before candidate ranking and keep the gate outside all generative agents.
WHY	Association, semantic similarity and plausible narrative do not establish causal relevance. Ranking without eligibility produces persuasive but scientifically weak candidates.
WHERE	The gate consumes colocalisation, fine-mapping/L2G, Mendelian and functional evidence from snapshot-bound feature views.
WHEN	Execute after endotype proposal and before taxonomy-dependent intervention scoring; re-run whenever causal source or calibration changes.
Execution mechanics	Policy defines minimum independent axes, evidence quality, context applicability, calibration and insufficiency codes. Output is PASS, FAIL or INDETERMINATE with references.
Rejected alternative	A single blended relevance score was rejected because high semantic or omics features could compensate for absent causal support.
Failure / degraded mode	FAIL remains hypothesis-only; INDETERMINATE enters an evidence-acquisition queue; no agent or human UI can directly rewrite the result.
Validation evidence	Validate known positives/negatives, correlated-source sensitivity, calibration, ancestry/tissue subgroups and attempted override.
Revisit trigger	Revisit thresholds when prospective validation changes the relationship between evidence axes and success.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-16 Abstaining direction resolver

Dimension	Principal-architecture position
Decision	Represent intervention direction as ACTIVATE, INHIBIT or UNKNOWN with basis, confidence and conflicts.
WHY	Genetic association identifies relevance but may not define therapeutic polarity. Wrong-direction promotion is a high-severity scientific failure.
WHERE	Resolver combines target action, signed networks, Mendelian loss/gain-of-function, perturbation and approved-precedent evidence in the model-and-gate tier.
WHEN	Run after candidate generation and before direction-dependent ranking; repeat for each tissue, cell context, dose and mechanism version.
Execution mechanics	Rules establish authoritative signs; graph propagation is bounded and uncertainty-aware; conflicting sources force UNKNOWN rather than majority coercion.
Rejected alternative	Binary sign inference and LLM interpretation were rejected because they erase conflict and create false certainty.
Failure / degraded mode	UNKNOWN blocks claims that depend on polarity but permits magnitude-only exploratory analysis with explicit labelling.
Validation evidence	Maintain a direction sentinel suite, conflict fixtures, path-sign tests and prospective outcome analysis by confidence band.
Revisit trigger	Revisit evidence precedence when pharmacology or genetic-direction standards change.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

DE-17 Post-review deterministic safety gate

Dimension	Principal-architecture position
Decision	Execute safety policy after all model ranking and reviewer adjustment, and make its verdict non-overridable by agents.
WHY	A reviewer can be persuaded by mechanistic benefit; safety asymmetry requires an independent final barrier that cannot be compensated by score.
WHERE	The gate reads labels, contraindications, interactions, adverse-event evidence, exposure, combination rules and dependency health.
WHEN	Evaluate immediately before evidence-card release and again when safety sources, candidate combination or intended population changes.
Execution mechanics	Signed rule bundles return PASS, BLOCK or INDETERMINATE with exact rule hits; unavailable critical dependencies fail closed.
Rejected alternative	Safety as a rank feature and pre-review-only filtering were rejected because later promotion could reintroduce blocked candidates.
Failure / degraded mode	BLOCK is immutable for that rule version; INDETERMINATE routes to safety review and cannot surface as eligible.
Validation evidence	Use sentinel cases, mutation testing, dependency outage drills, attempted agent override and rule-version replay.
Revisit trigger	Revisit a rule only through controlled scientific and safety change with impact analysis of prior decisions.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-18 Monotone learning-to-rank with decomposed trace

Dimension	Principal-architecture position
Decision	Use a constrained ranking model whose directionally known risk and evidence features obey monotonicity, while persisting feature contributions and rule traces.
WHY	Candidate ordering benefits from learned interactions, but scientifically nonsensical gradients undermine trust and can exploit data artifacts.
WHERE	SageMaker trains family-specific or hierarchical rankers over point-in-time pair features; batch inference materialises rankings into the snapshot.
WHEN	Train after label maturity and dataset audit; use on gate-passed candidates only; retrain on approved trigger, not every new result.
Execution mechanics	Group and temporal splits prevent target/scaffold leakage; constraints encode safety/risk/evidence monotonicity; SHAP or equivalent decomposition is stored with the decision.
Rejected alternative	Unconstrained deep ranking and hand-weighted scoring alone were rejected: one is hard to control, the other cannot learn validated nonlinear interactions.
Failure / degraded mode	Model failure falls back to the approved rule-based baseline; rank outputs without calibrator are not promoted.
Validation evidence	Evaluate NDCG/enrichment, stability, subgroup performance, monotonicity violations, calibration interaction and prospective hit rate.
Revisit trigger	Revisit architecture when labels support more expressive models without losing control or explanation.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-19 Separate ranking from calibrated uncertainty

Dimension	Principal-architecture position
Decision	Attach calibrated probability/utility estimates and conformal lower bounds rather than treating rank score as confidence.
WHY	Ordering objectives are not probabilities. Human triage needs a conservative measure of what is known, not merely which candidate scored highest.
WHERE	Calibration artifacts are versioned per model family, evidence stratum and relevant subgroup in SageMaker Model Registry.
WHEN	Fit only on held-out, point-in-time data; apply after rank inference and before reviewer promotion or portfolio selection.
Execution mechanics	Use isotonic/Platt where validated and split or Mondrian conformal methods for coverage; persist calibration cohort and interval method.
Rejected alternative	Softmax-normalised rank scores were rejected because they create confidence without empirical coverage.
Failure / degraded mode	Calibration drift demotes automatic promotion and widens/withholds uncertainty claims; base ranking may remain exploratory.
Validation evidence	Monitor ECE/Brier, interval coverage/width, subgroup undercoverage and decision quality by lower-bound threshold.
Revisit trigger	Revisit stratification when cohort size is insufficient or prospective outcomes reveal non-exchangeability.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-20 Value-of-information triage

Dimension	Principal-architecture position
Decision	Prioritise human review and wet-lab validation by expected information gain, portfolio relevance, uncertainty and experiment cost—not raw model score alone.
WHY	Validation capacity is the binding resource. Testing only top confidence reinforces existing beliefs and fails to improve uncertain model regions.
WHERE	The portfolio service consumes candidate utility, uncertainty, mechanism diversity, assay feasibility, cost, dependencies and strategic constraints.
WHEN	Run after safety and before validation planning; recompute when budget, assay capacity or evidence changes.
Execution mechanics	Optimisation produces a transparent queue with objective components, constraints, scenario sensitivity and human override reason codes.
Rejected alternative	Highest-score-first and autonomous Bayesian optimisation were rejected because they ignore portfolio concentration, feasibility and accountable strategy.
Failure / degraded mode	If optimisation is unavailable, use a deterministic stratified policy preserving uncertainty and mechanism diversity.
Validation evidence	Measure information gain, validation yield, diversity, budget adherence and downstream calibration improvement.
Revisit trigger	Revisit objective weights during quarterly portfolio governance or when programme strategy changes.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-21 Point-in-time dataset compiler

Dimension	Principal-architecture position
Decision	Generate every training and evaluation row from a declared as-of cut-off, source availability lag and snapshot manifest.
WHY	Biomedical data has publication and curation delays; naive timestamp filters still allow future knowledge and duplicated evidence into historic decisions.
WHERE	The compiler joins Iceberg tables, feature history, retrieval corpora, ontology versions, foundation-model cut-offs and adjudicated labels.
WHEN	Run for every experiment, benchmark refresh and regulatory validation set; freeze the manifest before model fitting.
Execution mechanics	Entity, target, scaffold, publication lineage and time partitions are applied before feature materialisation; late labels remain unavailable until maturity.
Rejected alternative	Random row splitting and latest-corpus retrieval were rejected because performance would measure leakage rather than generalisation.
Failure / degraded mode	Any unresolved lineage or post-cut-off evidence invalidates the dataset build and blocks training.
Validation evidence	Use leakage canaries, duplicate/lineage analysis, reconstruction from manifest and independent query review.
Revisit trigger	Revisit lag assumptions when source publishing or internal adjudication processes change.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-22 Grouped temporal scientific evaluation

Dimension	Principal-architecture position
Decision	Evaluate with temporal and biological grouping—target, scaffold, disease family, assay lineage or partner—selected to match the deployment claim.
WHY	Random splits overstate performance when related molecules, genes, papers or assays appear on both sides.
WHERE	Applied to causal models, rankers, extractors, embeddings, safety models and generative benchmark cohorts.
WHEN	Choose before training based on intended generalisation; report multiple splits when the model serves more than one novelty regime.
Execution mechanics	An evaluation plan declares unit of independence, cut-off, groups, exclusions, subgroup minimums, metric confidence intervals and acceptance thresholds.
Rejected alternative	One aggregate benchmark was rejected because it conceals leakage and underperformance in rare or novel strata.
Failure / degraded mode	A model failing the deployment-matched split remains experimental even if its random-split metric is high.
Validation evidence	Audit group overlap, bootstrap by independent unit, compare masked/unmasked retrieval and perform prospective shadow evaluation.
Revisit trigger	Revisit grouping when product scope or biological novelty claim changes.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-23 Validation outcome as context-rich label

Dimension	Principal-architecture position
Decision	Represent laboratory and translational outcomes with protocol, assay, model system, dose, exposure, endpoint, QC, deviation and adjudication—not a bare binary label.
WHY	A result is only meaningful in experimental context. Collapsing outcomes creates label noise and unsafe transfer across assays or biological settings.
WHERE	Validation packages and results live in tenant/partner-isolated stores with immutable raw-data references and governed feature projections.
WHEN	Create a training-eligible label only after protocol completion, QC and scientific adjudication; preserve inconclusive and failed-assay states.
Execution mechanics	The label schema links hypothesis, protocol version, materials, run, controls, raw data, effect/uncertainty, deviations, signer and maturity.
Rejected alternative	Spreadsheet summaries and automatic positive/negative conversion were rejected because chain of custody and assay validity disappear.
Failure / degraded mode	Quarantined results never update models; conflicting valid runs remain separate and trigger adjudication rather than overwrite.
Validation evidence	Reconcile package-to-result manifests, repeatability, QC, signer identity, label agreement and downstream sensitivity.
Revisit trigger	Revisit label abstraction when sufficient evidence supports validated cross-assay harmonisation.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-24 Model, data, prompt, rule and agent registries

Dimension	Principal-architecture position
Decision	Treat every executable or policy-bearing AI artifact as independently versioned, owned, evaluated and rollback-capable.
WHY	A scientific output depends on more than model weights. Prompt, retrieval, rule, schema and agent changes can alter decisions without code release.
WHERE	SageMaker Model Registry/managed MLflow track models; dedicated registries track datasets, prompts, schemas, routes, tools, agents and rule bundles.
WHEN	Register at build; promote only after linked evaluation and approvals; resolve versions once per snapshot or controlled online route.
Execution mechanics	Each artifact records intended use, owner, lineage, digest, compatibility, evaluation, effective interval, dependencies and rollback target.
Rejected alternative	Git commit alone and latest aliases were rejected because runtime composition and approval evidence are missing.
Failure / degraded mode	Unresolvable or incompatible artifact versions fail compilation; serving never silently upgrades.
Validation evidence	Periodically select released decisions and traverse every artifact and approval to source evidence.
Revisit trigger	Revisit registry boundaries when a new artifact can independently change scientific behaviour.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-25 Retrieval policy before similarity scoring

Dimension	Principal-architecture position
Decision	Apply identity, tenant, licence, date, intended-use and publication-state filters before vector or lexical scoring.
WHY	Post-filtering can expose forbidden content to ranking, caching, traces or model context even if the final response hides it.
WHERE	Policy compilation sits in front of OpenSearch, Neptune and evidence APIs; filters are included in query and cache identity.
WHEN	Execute on every retrieval hop and graph expansion, including agent-initiated follow-ups.
Execution mechanics	The policy decision emits an allow-set or signed predicate; retrieval services reject queries without it and return references with applied-policy hash.
Rejected alternative	Retrieve-all-then-redact was rejected because sensitive content had already crossed the boundary.
Failure / degraded mode	Policy service failure denies retrieval; approved deterministic templates and public evidence may remain available under explicit policy.
Validation evidence	Use synthetic forbidden documents, cross-tenant probes, licence changes, deletion tests and trace inspection.
Revisit trigger	Revisit indexing topology if mandatory pre-filtering degrades recall or latency beyond the service objective.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

DE-26 Hybrid retrieval with contradiction quota

Dimension	Principal-architecture position
Decision	Retrieve structured facts, lexical passages, vectors and bounded graph paths, and reserve context capacity for negative or contradictory evidence.
WHY	Similarity systems preferentially return confirming material. Scientific review requires disconfirming evidence and source diversity to avoid narrative lock-in.
WHERE	The context compiler merges Aurora, OpenSearch and Neptune candidates and labels support, contradiction, applicability and source tier.
WHEN	Use for hypothesis authoring, reviewer promotion, mechanism explanation and regulatory drafting; simpler exact lookups bypass semantic retrieval.
Execution mechanics	Fusion and reranking preserve channel scores; contradiction queries use claim predicates and negation/adverse outcome patterns; diversity constraints cap one source.
Rejected alternative	Pure top-k semantic retrieval was rejected because it suppresses rare negative results and duplicates popular sources.
Failure / degraded mode	If contradiction retrieval is incomplete, the artifact declares the gap and cannot claim comprehensive review.
Validation evidence	Evaluate contradiction recall, evidence diversity, channel ablations, claim support and reviewer decisions with/without negative evidence.
Revisit trigger	Revisit quotas by task when they reduce relevant support or fail to surface known counter-evidence.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

DE-27 Bounded GraphRAG path templates

Dimension	Principal-architecture position
Decision	Use predefined typed path shapes and hop/node/time budgets for causal, mechanism, biomarker, safety, validation and regulatory trace questions.
WHY	Unbounded graph exploration is expensive, vulnerable to poisoning/path explosion and hard to evaluate. Domain path semantics can be declared explicitly.
WHERE	Templates execute in Neptune with tenant/effective-date predicates and hydrate each edge to source evidence before entering context.
WHEN	Use when the question requires relationship composition; direct facts and high-volume analytics use relational or offline graph outputs.
Execution mechanics	A template declares allowed nodes/edges, sign rules, maximum hops, context filters, ranking and required hydration. Results store query-plan hash.
Rejected alternative	Model-generated Cypher/openCypher was rejected for production because cost, access and semantic constraints are not predictable.
Failure / degraded mode	Timeout or missing hydration yields incomplete path status; the system does not invent a bridging relation.
Validation evidence	Maintain path gold sets, mutation tests, budget tests, poisoning corpora and source-hydration precision.
Revisit trigger	Revisit templates from observed analyst queries through architecture review, never autonomous generation.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

DE-28 Tenant isolation at every persistence and compute layer

Dimension	Principal-architecture position
Decision	Carry tenant identity from federated token through workflow, storage keys, query predicates, caches, model context, trace and cost ledger.
WHY	Application-only isolation fails when one downstream service omits a filter. Defence in depth makes a single defect insufficient for disclosure.
WHERE	Identity, Aurora RLS, S3 prefixes/access points, Lake Formation, KMS context, OpenSearch/Neptune policy, queues, feature keys and AgentCore sessions enforce scope.
WHEN	Bind at request admission and propagate immutably; background jobs use explicit tenant sets rather than inferred context.
Execution mechanics	Policy-as-code tests verify principal, tenant, purpose, resource and operation. Large or regulated partners can receive dedicated account/cell resources.
Rejected alternative	ORM filters and shared unqualified caches were rejected because bypass is easy and evidence of isolation is weak.
Failure / degraded mode	Missing tenant context is a hard deny; suspected leakage activates route/snapshot/tenant kill switches and incident workflow.
Validation evidence	Run cross-tenant synthetic probes, cache-collision tests, confused-deputy attacks and data-lineage scans.
Revisit trigger	Revisit shared versus dedicated placement using risk, noisy-neighbour and total-cost evidence.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

DE-29 Encryption context and cryptographic deletion

Dimension	Principal-architecture position
Decision	Use KMS keys and encryption context aligned to environment, data class and tenant; support crypto-shredding where contractual deletion requires it.
WHY	Encryption at rest is insufficient if broad principals can decrypt across partners. Key policy is an authorisation and lifecycle control.
WHERE	S3, Aurora, OpenSearch, Neptune exports, queues, feature stores, validation packages and backups use managed encryption with controlled grants.
WHEN	Assign context at object/record creation; rotate by policy; schedule deletion only after retention, legal-hold and dependency checks.
Execution mechanics	Workloads receive short-lived decrypt grants bound to purpose and resource; key use is audited centrally.
Rejected alternative	One platform-wide key was rejected because compromise and offboarding blast radius were unacceptable.
Failure / degraded mode	KMS outage blocks protected writes/reads according to capability tier; released public artifacts may remain available.
Validation evidence	Test wrong-context decrypt, grant expiry, rotation, restore, legal hold and verifiable crypto-shred workflow.
Revisit trigger	Revisit key granularity when key-count operations exceed value or partner contract demands stronger separation.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-30 No shared semantic cache for confidential evidence

Dimension	Principal-architecture position
Decision	Disable approximate semantic caching for partner compounds, validation data and sensitive scientific prompts; use exact, tenant-qualified caches where safe.
WHY	Semantic keys can collide across similar confidential hypotheses and disclose both content and the existence of another partner programme.
WHERE	Cache policy is enforced at the Bedrock gateway, retrieval layer and tool gateway using tenant, snapshot, purpose, data class and artifact versions.
WHEN	Decide before lookup and insertion; confidential contexts bypass semantic caches entirely.
Execution mechanics	Exact caches store references to encrypted artifacts, short TTL and policy hash. Public non-sensitive narration may use evaluated semantic caching.
Rejected alternative	A global response cache was rejected despite cost savings because approximate reuse violates isolation and freshness guarantees.
Failure / degraded mode	Cache-service failure bypasses cache rather than serving an unverified entry; correctness and confidentiality dominate latency.
Validation evidence	Use collision attacks, tenant substitution, stale-version tests, deletion propagation and cache-key inspection.
Revisit trigger	Revisit only if a privacy-preserving tenant-local cache demonstrates measurable benefit and passes threat review.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-31 Immutable audit and decision lineage

Dimension	Principal-architecture position
Decision	Record model, tool, retrieval, policy, human, validation and release events in an immutable audit stream linked by trace and decision identifiers.
WHY	Regulatory readiness requires proof of what happened, not reconstructed application logs. Scientific replay also needs event order and artifact identity.
WHERE	CloudTrail, application audit events, Step Functions history, AgentCore traces and decision records converge into Security Lake/S3 Object Lock with searchable projections.
WHEN	Emit before/after every privileged action, artifact promotion, policy decision, human approval, external transfer and kill-switch operation.
Execution mechanics	Events are schema-versioned, time-synchronised, hash-linked where required, encrypted and retained by record class; payloads use secure references to minimise sensitive duplication.
Rejected alternative	Mutable operational logs were rejected because administrators could alter or lose evidence and retention was inconsistent.
Failure / degraded mode	If audit persistence is unavailable, regulated mutations stop; read-only serving may continue with local buffering only where approved.
Validation evidence	Reconcile workflow history to audit events, verify retention/object lock, test signer identity and replay sampled decisions.
Revisit trigger	Revisit event payload and retention when regulatory classification or investigation needs change.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-32 Electronic approval as a domain primitive

Dimension	Principal-architecture position
Decision	Model scientific, quality, safety and regulatory approvals as identity-bound signed records with scope, reason, evidence hash and expiry.
WHY	A UI button or ticket comment is not sufficient evidence of accountable review and can approve an artifact that later changes.
WHERE	Approval records live in Aurora and the immutable audit archive; Step Functions gates release and validation transitions on exact artifact hashes.
WHEN	Require at risk-defined transitions: snapshot release, model/rule promotion, safety exception process, protocol approval and dossier issuance.
Execution mechanics	Approver role and separation-of-duty policy are checked at signature time; edits invalidate the prior approval and create a new version.
Rejected alternative	Informal email and generic change-ticket approval were rejected because scope and evidence identity were ambiguous.
Failure / degraded mode	Unavailable approval service pauses the transition; no cached approval applies to a changed artifact.
Validation evidence	Test stale approval, self-approval, delegated authority, expiry, evidence mutation and audit reconstruction.
Revisit trigger	Revisit approval matrix through quality governance when risk classification or organisation roles change.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-33 Risk-tiered multi-Region resilience

Dimension	Principal-architecture position
Decision	Assign RTO/RPO by capability: safety and released evidence before interactive retrieval, and interactive retrieval before compilation/training.
WHY	Uniform active-active design is expensive and scientifically unnecessary. The critical requirement is safe, traceable decision continuity, not uninterrupted generation.
WHERE	Replicate signed S3 artifacts, backups and required databases cross-Region; provision warm or pilot-light Neptune/OpenSearch/compute based on tier.
WHEN	Continuously replicate C0/C1 data; exercise failover under change control; activate secondary only after signature and isolation probes.
Execution mechanics	A continuity workflow freezes promotion, verifies replication, changes aliases/routes, executes canaries, restores queues and reconciles work.
Rejected alternative	Full active-active for every pipeline was rejected because conflict resolution and cost exceed the value of continuous batch compilation.
Failure / degraded mode	During regional loss, serve prior released snapshots and deterministic safety/ranking; defer generative authoring and new training if required.
Validation evidence	Measure real game-day RTO/RPO, decision correctness, tenant isolation and reconciliation—not design estimates.
Revisit trigger	Revisit tiers after material business-impact, validation-throughput or regulatory change.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-34 Graceful AI degradation

Dimension	Principal-architecture position
Decision	Design every probabilistic capability with a safer fallback: deterministic template, base rank, exact retrieval, human queue or prior snapshot.
WHY	Foundation models and vector/graph services will fail or regress. Safety and evidence access must not depend on their availability.
WHERE	Circuit breakers at the gateway and workflow route by capability, model family, Region and dependency health.
WHEN	Activate on error/latency burn rate, guardrail anomaly, grounding regression, budget exhaustion or operator kill switch.
Execution mechanics	Fallback status is visible in decision records and user interfaces; degraded artifacts cannot claim the same completeness or autonomy.
Rejected alternative	Retry-until-success and silent model substitution were rejected because they create latency cascades and unvalidated behaviour.
Failure / degraded mode	Safety dependencies fail closed; narration may template; reviewer failure yields no promotion; graph failure yields structured-only evidence.
Validation evidence	Chaos-test each dependency and assert output authority, user labelling, no duplicate effects and recovery criteria.
Revisit trigger	Revisit fallback hierarchy after every incident and model-route evaluation.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-35 Decision-level FinOps ledger

Dimension	Principal-architecture position
Decision	Attribute compute, model, retrieval, graph, storage, validation and human-review cost to snapshot, tenant, capability, hypothesis and decision.
WHY	Service-level cloud spend cannot answer whether the architecture economically produces validated scientific value.
WHERE	Usage events from Bedrock, SageMaker, EMR, Neptune, OpenSearch, workflows and validation operations feed a governed cost semantic layer.
WHEN	Estimate before expensive calls/jobs; record actual after completion; aggregate at release, portfolio and partner review.
Execution mechanics	The ledger includes tokens, accelerator time, scanned bytes, graph capacity, storage, retries, human time and external validation charges.
Rejected alternative	Monthly account totals were rejected because optimisation could not identify which decision path or artifact created cost.
Failure / degraded mode	Budget breach downgrades model tier, narrows optional retrieval or pauses non-critical compilation; never weakens safety or evidence controls.
Validation evidence	Reconcile tagged usage to billing, detect unattributed spend, compare estimate to actual and measure cost per accepted/validated hypothesis.
Revisit trigger	Revisit allocation keys when product pricing or organisational accountability changes.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-36 Denial-of-wallet protection

Dimension	Principal-architecture position
Decision	Enforce call, token, retrieval, graph, wall-clock, concurrency and spend budgets in runtime policy rather than prompt text.
WHY	Agent loops and adversarial queries can generate unbounded cost even when individual calls are valid. Cost exhaustion is also an availability threat.
WHERE	Budgets are carried in workflow/agent state and checked by Bedrock gateway, AgentCore tools, OpenSearch/Neptune query wrappers and batch schedulers.
WHEN	Preflight before each operation and decrement from actual usage; terminate on cycle/state-hash recurrence or deadline.
Execution mechanics	Policy varies by tenant, capability and phase; hard safety queries have reserved capacity and cannot be starved by generative workloads.
Rejected alternative	Prompted requests to be concise and account-only budgets were rejected because they do not stop one pathological trajectory.
Failure / degraded mode	On budget exhaustion, return PARTIAL with omitted work and route to human or later batch; never portray partial review as complete.
Validation evidence	Red-team graph explosion, recursive repair, high-k retrieval, oversized documents and concurrent tenant bursts.
Revisit trigger	Revisit limits from p95/p99 telemetry and validated quality-versus-cost curves.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-37 Prompt-injection defence outside the model

Dimension	Principal-architecture position
Decision	Treat retrieved documents, tool outputs and user text as untrusted data; enforce authority, tool policy and output validation outside foundation models.
WHY	No prompt can guarantee that another prompt is ignored. Biomedical documents can contain accidental or malicious instructions and encoded exfiltration attempts.
WHERE	The context compiler labels channels; AgentCore Gateway constrains tools; egress policy, DLP, canary secrets and deterministic validators run outside Bedrock.
WHEN	Evaluate before tool discovery, after tool arguments, after retrieval, before external egress and before state commit.
Execution mechanics	Use content normalisation, instruction stripping where safe, suspicious-span tagging, capability tokens, result sanitisation and independent policy checks.
Rejected alternative	A stronger system prompt alone was rejected because model obedience is probabilistic and changes by version.
Failure / degraded mode	Suspicious context reduces tool authority, quarantines output or routes to human; critical workflows continue with structured facts/templates.
Validation evidence	Maintain direct/indirect, multi-turn, OCR, encoding, tool-output and graph-poisoning corpora across all model routes.
Revisit trigger	Revisit controls after each model upgrade, new document channel or red-team finding.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

DE-38 Scientific data-poisoning controls

Dimension	Principal-architecture position
Decision	Authenticate source releases, preserve checksums, detect distribution/relationship anomalies and require source diversity for high-impact evidence.
WHY	A valid schema does not imply truthful evidence. Poisoned or corrupted source data can influence graph paths, features and model labels.
WHERE	Controls operate at S3 landing, harmonisation, graph publication, feature compilation, retrieval indexing and validation-result ingestion.
WHEN	Run on each source release and before snapshot promotion; re-run targeted impact analysis after a source correction.
Execution mechanics	Compare publisher signatures/checksums, historical statistics, entity/edge anomalies, duplicates, contradiction changes and unexpected concentration by source.
Rejected alternative	Trusting upstream availability and relying only on model robustness were rejected because poisoned evidence would still appear in explanations.
Failure / degraded mode	Quarantine affected source partitions and build an explicitly incomplete snapshot only with scientific approval; prior release remains available.
Validation evidence	Use seeded corruptions, source-removal sensitivity, graph motif anomalies and decision-impact traversal.
Revisit trigger	Revisit thresholds when legitimate source methodology materially changes distributions.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-39 Licence-aware derivation and release packaging

Dimension	Principal-architecture position
Decision	Attach licence and permitted-use metadata to source objects, transformed rows, graph edges, features and artifacts; compile releases through a licence policy gate.
WHY	Biomedical sources have incompatible attribution, redistribution and commercial-use terms. Technical lineage must support legal policy at derivative granularity.
WHERE	Lake Formation tags, catalog metadata, decision lineage and release packager separate open, proprietary, restricted and partner-derived products.
WHEN	Evaluate at ingestion, transformation, training, retrieval, partner export and public release; changes trigger impact analysis.
Execution mechanics	Derived artifacts carry contributing-source sets and policy class; packager builds only from an allow-list and emits attribution/obligation manifest.
Rejected alternative	A document-level licence spreadsheet was rejected because obligations disappear after joins, embeddings and feature construction.
Failure / degraded mode	Unknown or incompatible rights block export/training for the intended use but may remain quarantined for review.
Validation evidence	Test prohibited-source canaries, derivative lineage, model/dataset manifests, deletion and release contents.
Revisit trigger	Revisit policy rules with counsel when source terms, jurisdiction or product intended use changes.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-40 Deletion across derived AI artifacts

Dimension	Principal-architecture position
Decision	Implement deletion as lineage-driven propagation across source, chunks, vectors, graph edges, features, datasets, caches and tenant adapters.
WHY	Removing an operational row does not remove memorised or indexed derivatives. Partner rights and privacy commitments require verifiable end-to-end erasure.
WHERE	A deletion orchestrator traverses catalog/lineage metadata and invokes S3, Aurora, OpenSearch, Neptune, feature, cache and model-governance actions.
WHEN	Trigger on authorised request, source withdrawal, licence expiry or partner offboarding; respect legal hold and immutable regulated records.
Execution mechanics	Each task is idempotent, evidenced and reconciled; models are retrained or risk-assessed when exact unlearning is unavailable.
Rejected alternative	Best-effort asynchronous deletion without a completion proof was rejected because invisible derivatives persist.
Failure / degraded mode	Failed tasks keep the request open and block affected artifacts from future release until resolved.
Validation evidence	Seed traceable synthetic records and prove disappearance from every serving and training path, including backups according to policy.
Revisit trigger	Revisit unlearning strategy when tenant adaptation or training frequency changes.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-41 Regulatory evidence assembled from the decision graph

Dimension	Principal-architecture position
Decision	Generate controlled dossier components by traversing exact requirements, sources, features, models, rules, reviews, validation results and changes—not by asking an LLM to reconstruct history.
WHY	Narrative fluency is not traceability. Regulatory readiness depends on stable evidence identity and demonstrable design/verification linkage.
WHERE	The regulatory evidence service reads immutable decision records, audit events, registries, validation packages and controlled templates.
WHEN	Assemble for formal review, partner due diligence, change assessment and submission; freeze against one released snapshot and intended use.
Execution mechanics	Deterministic assembly creates the traceability matrix and evidence tables; Bedrock may draft prose only from those records and citations.
Rejected alternative	Free-form dossier generation and manual copy/paste were rejected because omissions and version mixing are difficult to detect.
Failure / degraded mode	Missing evidence produces an explicit gap, not inferred text; controlled issuance waits for QA/RA approval.
Validation evidence	Perform bidirectional requirement trace, citation/source existence, version-consistency and mock-audit replay.
Revisit trigger	Revisit templates by jurisdiction and intended use while retaining the same decision graph.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

DE-42 Controlled change impact graph

Dimension	Principal-architecture position
Decision	Compute the impact of source, ontology, mapping, feature, graph, model, prompt, rule or schema change on released decisions and dossier artifacts.
WHY	AI systems have non-local dependencies. A small mapping or prompt change can alter many scientific conclusions without an obvious code diff.
WHERE	Registry dependency edges and decision lineage feed a change-impact service used by quality, model risk and release workflows.
WHEN	Run at change proposal, before validation scope approval, after implementation and before release.
Execution mechanics	Impact output lists affected snapshots, entities, decisions, validation packages, metrics, documents and required re-evaluation; owners approve residual risk.
Rejected alternative	Uniform full revalidation and developer judgement alone were rejected as respectively wasteful and unreliable.
Failure / degraded mode	Unknown lineage expands the scope conservatively or blocks the change; it never shrinks required validation.
Validation evidence	Compare predicted impact with shadow-snapshot diff and audit false-negative impact misses.
Revisit trigger	Revisit dependency granularity when impact sets remain systematically too broad or miss observed changes.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-43 Artifact-level progressive delivery

Dimension	Principal-architecture position
Decision	Canary models, prompts, retrieval indexes, agent graphs and rule bundles independently while binding the tested combination in a candidate snapshot.
WHY	A single application deployment hides which AI artifact changed behaviour and makes rollback unnecessarily broad.
WHERE	Aliases and route policy in Bedrock, SageMaker, OpenSearch, Neptune projections and registries support shadow, cohort and full promotion.
WHEN	Use after frozen replay and before release; select non-regulated or mandatory-review cohorts according to risk.
Execution mechanics	Collect paired decisions, quality, safety, cost and latency; block cross-artifact incompatibility through manifest constraints.
Rejected alternative	Big-bang deployment and untracked online experimentation were rejected because affected decisions and rollback targets were ambiguous.
Failure / degraded mode	A failing artifact alias reverts independently; if interaction is suspected, revert the entire candidate snapshot.
Validation evidence	Use sequential tests or confidence intervals appropriate to cohort size, plus zero-tolerance gate metrics.
Revisit trigger	Revisit canary duration and cohort allocation from incident and drift evidence.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-44 Scientific SLOs over generic uptime

Dimension	Principal-architecture position
Decision	Define SLOs for decision trace completeness, hard-gate integrity, citation support, direction error, calibration coverage, validation lineage and replay—not only API latency.
WHY	A highly available system can still be scientifically unsafe. SRE must observe correctness and evidence integrity as production reliability dimensions.
WHERE	Metrics derive from decision records, gate events, retrieval checks, validation lineage and workflow traces; CloudWatch hosts burn-rate alarms.
WHEN	Evaluate continuously for serving and per snapshot for batch; page on fast burn for safety/isolation and ticket slower quality decay.
Execution mechanics	Error budgets determine promotion, autonomy demotion and engineering priority. Denominators and exclusions are registry-controlled.
Rejected alternative	Generic 99.9% availability alone was rejected because it ignores unsupported or wrong-direction outputs.
Failure / degraded mode	Quality-budget exhaustion freezes promotion, lowers autonomy or routes to human while retaining safe read access.
Validation evidence	Inject synthetic sentinels, reconcile metrics to raw decisions and review error-budget policy quarterly.
Revisit trigger	Revisit targets when scientific evidence supports a tighter bound or product intended use changes.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-45 Prospective shadow validation before autonomy

Dimension	Principal-architecture position
Decision	Require prospective shadow operation and mandatory human review before an AI capability changes scientific workflow automatically.
WHY	Retrospective performance cannot capture distribution shift, user adaptation, operational latency or hidden data-quality dependencies.
WHERE	Shadow outputs are produced in the live pipeline but isolated from committed state; reviewers compare them with actual decisions and outcomes.
WHEN	Run after offline release gates for each new model/capability, material data expansion or autonomy increase.
Execution mechanics	Define sample, duration, strata, disagreement adjudication, stop conditions and success thresholds before launch; preserve counterfactual decisions.
Rejected alternative	Immediate auto-execution after benchmark success was rejected because offline metrics do not prove workflow safety.
Failure / degraded mode	Any gate violation stops shadow and opens incident/change review; no partial autonomy is granted by the model team alone.
Validation evidence	Measure agreement, beneficial disagreement, unsupported claims, latency, cost, reviewer burden and prospective outcomes.
Revisit trigger	Revisit autonomy only through governance after sufficient shadow evidence.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-46 Human review as training-data product

Dimension	Principal-architecture position
Decision	Design review UI and reason codes jointly with model and data teams so edits, vetoes and overrides become interpretable labels.
WHY	Unstructured approval clicks do not explain whether the model was wrong, evidence was missing or policy was conservative.
WHERE	Review events link proposal, evidence, model/prompt/rule versions, structured defects, edit patch, decision and downstream outcome.
WHEN	Capture at every required interrupt; use labels only after quality checks and consent/contract policy.
Execution mechanics	Separate factual correction, preference, policy, safety, novelty and workflow reasons; retain reviewer role and confidence.
Rejected alternative	Mining free-text comments alone was rejected because labels are inconsistent and hard to audit.
Failure / degraded mode	Low-agreement categories remain evaluation-only until adjudication; no automatic retraining from raw clicks.
Validation evidence	Monitor reviewer agreement, edit distance, reason completeness, position bias and label drift.
Revisit trigger	Revisit taxonomy when reviewers overuse other or when model failure modes evolve.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-47 No partner data in shared training by default

Dimension	Principal-architecture position
Decision	Exclude partner compounds, assays, messages and validation outcomes from global training unless an explicit contractual, consent and governance path permits it.
WHY	Inference permission does not imply training permission; model memorisation and cross-partner benefit create legal and trust risk.
WHERE	Dataset compiler requires a training-use entitlement distinct from storage and inference access; tenant adapters remain tenant-scoped.
WHEN	Evaluate at ingestion and every dataset build; entitlement changes apply prospectively through controlled rebuild.
Execution mechanics	Dataset manifests include source/tenant rights, de-identification, retention, purpose and exclusion evidence.
Rejected alternative	Opt-out filtering after dataset creation was rejected because copied features and checkpoints are hard to unwind.
Failure / degraded mode	Unclear entitlement is exclusion, not assumption; model quality degrades transparently rather than borrowing unauthorised data.
Validation evidence	Use prohibited-tenant canaries, lineage queries, membership-inference tests and manifest review.
Revisit trigger	Revisit only through legal/data-governance approval and documented value/risk assessment.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-48 Custom training on SageMaker with reproducible recipes

Dimension	Principal-architecture position
Decision	Use SageMaker Pipelines/Training/Processing, managed MLflow and Model Registry for causal, ranking, calibration, extraction and embedding models.
WHY	Custom models are justified only when stable task volume, proprietary labels, latency, privacy or cost beats managed foundation-model routing.
WHERE	Training jobs run in isolated VPC subnets over manifest-bound S3/Iceberg datasets with signed containers and no public egress.
WHEN	Train on approved triggers: enough new mature labels, measured drift, source-domain expansion or algorithmic change—not a calendar alone.
Execution mechanics	Pipeline stages validate data, build PIT splits, train, calibrate, evaluate, scan, register and produce model/data cards and deployment package.
Rejected alternative	Notebook-to-endpoint and automatic continuous retraining were rejected because evidence, cost and change control are insufficient.
Failure / degraded mode	Failed stages leave immutable artifacts but no promotable model; previous registry version remains served.
Validation evidence	Re-execute recipe from manifest, compare deterministic seeds/tolerances, validate metrics, robustness, security and inference parity.
Revisit trigger	Revisit managed versus self-hosted economics per capability and planning load.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-49 Model distillation only after trace-quality proof

Dimension	Principal-architecture position
Decision	Distil high-volume extraction or narration tasks into smaller models only after a teacher route demonstrates stable, grounded behaviour and labels pass review.
WHY	Distillation can reduce cost and latency but also fossilise teacher errors and obscure uncertainty.
WHERE	SageMaker training uses approved teacher outputs, human corrections and hard negatives; distilled models serve narrow capability classes.
WHEN	Initiate when route volume and quality stability justify dataset creation; repeat after significant teacher/policy change.
Execution mechanics	Filter by grounding and reviewer acceptance, balance rare defects, retain provenance to teacher artifact and evaluate independently on human-labelled cohorts.
Rejected alternative	Blind synthetic-data training was rejected because confident teacher mistakes become training truth.
Failure / degraded mode	If drift or defect rate rises, router returns to managed model or deterministic baseline while new labels are adjudicated.
Validation evidence	Compare quality, calibration, robustness, subgroup performance, latency and total cost against teacher and baseline.
Revisit trigger	Revisit when foundation-model price/performance changes the crossover point.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-50 Safety and scientific rule bundles as code

Dimension	Principal-architecture position
Decision	Version deterministic scientific and safety policy as signed, tested rule bundles with rationale, evidence, owners and effective intervals.
WHY	Rules otherwise become hidden application branches or spreadsheets that cannot be replayed or independently approved.
WHERE	A rule service executes causal eligibility, direction precedence, safety veto, licence and release policies; bundles are referenced in every decision.
WHEN	Evaluate in compilation and request-time gates; deploy through shadow diff and two-person scientific/quality approval.
Execution mechanics	Rules use typed predicates and return exact hits; mutation tests and sentinel fixtures prove both positive and negative behaviour.
Rejected alternative	Embedding rules in prompts or UI code was rejected because authority and version identity disappear.
Failure / degraded mode	Invalid or unavailable bundles fail closed for governed transitions; prior bundle remains available for rollback.
Validation evidence	Run mutation coverage, boundary-value, conflict, performance, impact and historical replay tests.
Revisit trigger	Revisit rules when source standards, safety evidence or prospective validation changes.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

DE-51 Canonical scientific decision record

Dimension	Principal-architecture position
Decision	Persist one append-only record joining inputs, evidence, features, models, prompts, rules, gates, human decisions, validation and signatures for each surfaced hypothesis.
WHY	Without a canonical record, explanations, audit, retraining and regulatory packages reconstruct different versions of the truth.
WHERE	The record spans Aurora identity/state, S3 artifact references and immutable audit; APIs return record references rather than regenerate provenance.
WHEN	Create at first hypothesis, append state transitions and seal at release/validation decisions; corrections supersede rather than overwrite.
Execution mechanics	Hash-chained versions distinguish computed facts, generated proposals and human approvals; large payloads remain content-addressed by reference.
Rejected alternative	Logging model prompts alone and separate per-service audit tables were rejected because cross-stage causality was absent.
Failure / degraded mode	If record write fails, no governed side effect commits; reconciliation verifies event and record completeness.
Validation evidence	Sample end-to-end trace, reproduce deterministic fields, verify source/artifact existence and ensure no prohibited sensitive duplication.
Revisit trigger	Revisit schema when a new decision authority or evidence class enters the platform.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-52 CRO validation exchange as controlled supply chain

Dimension	Principal-architecture position
Decision	Treat protocol, materials, transfer, execution, deviation, raw results and acknowledgement as a signed scientific supply chain.
WHY	External validation is not a file upload; identity, chain of custody and protocol version determine whether results can update evidence.
WHERE	A partner integration account exposes signed package APIs, object transfer, event acknowledgement and isolated result ingestion.
WHEN	Create after human-approved validation plan; lock protocol before transfer; amendments create new versions and impact acceptance criteria.
Execution mechanics	Packages carry manifests, checksums, classification, materials/specimens, assays, endpoints, acceptance rules and responsible signers.
Rejected alternative	Email attachments and manually re-keyed spreadsheets were rejected because integrity and version reconciliation are weak.
Failure / degraded mode	Unexpected, unsigned or protocol-mismatched results enter quarantine; workflow preserves timeline and requests correction.
Validation evidence	Exercise duplicate/out-of-order delivery, tampered object, amendment, missing control, deviation and partner outage scenarios.
Revisit trigger	Revisit exchange protocol as partner automation maturity or regulatory obligations evolve.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-53 Regulatory-ready electronic records without over-claiming

Dimension	Principal-architecture position
Decision	Implement design control, audit, validation and electronic approvals where required while keeping research-use intended use explicit.
WHY	Premature clinical claims create regulatory exposure; absence of controlled engineering evidence blocks later qualification.
WHERE	Product claims, model cards, interfaces and dossier templates carry intended-use classification; quality controls scale by risk.
WHEN	Classify before development and re-evaluate on user, workflow, claim, autonomy, data or jurisdiction change.
Execution mechanics	A claims library, prohibited-use rules and change-impact workflow align technical outputs with approved wording and evidence maturity.
Rejected alternative	Either ignoring regulatory readiness or claiming full compliance by architecture alone was rejected; both are misleading.
Failure / degraded mode	Unclassified or expanded use is blocked from controlled release and routed to regulatory assessment.
Validation evidence	Mock audits, claim-to-evidence trace, electronic-record tests and validation-plan review assess readiness.
Revisit trigger	Revisit classification at every product-scope and partner-use expansion.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-54 Architecture roadmap gated by control maturity

Dimension	Principal-architecture position
Decision	Sequence evidence contracts and deterministic gates before GenAI autonomy, and validation closure before regulatory automation.
WHY	Building visible agents first creates technical debt in lineage, authority and evaluation. Controls introduced later cannot reconstruct missing history.
WHERE	The programme runs coordinated data/knowledge, ML, agent, science, quality, security/SRE and FinOps workstreams under one architecture runway.
WHEN	Advance phases only when exit evidence exists; parallelise implementation within a phase without bypassing dependency gates.
Execution mechanics	Each phase has architecture acceptance criteria, accountable owner, measurable evidence, risk register and rollback/de-scope option.
Rejected alternative	Feature-led roadmap and fixed-date autonomy commitments were rejected because they reward demo output over scientific control maturity.
Failure / degraded mode	If a gate slips, retain lower autonomy and prior snapshot; do not waive causal, safety, isolation or audit controls to protect a date.
Validation evidence	Architecture reviews verify exit artifacts, integration traces, game days, prospective shadow results and mock audit.
Revisit trigger	Revisit sequencing when external validation or regulatory deadlines change, while preserving hard dependencies.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

Jyotirmoy Bardhan

DE-55 Principal architecture veto and exception process

Dimension	Principal-architecture position
Decision	Assign explicit veto authority for causal, safety, security, tenant-isolation, lineage and unrecoverable architecture violations; manage exceptions as expiring signed risks.
WHY	Cross-functional programmes otherwise convert critical controls into negotiable backlog items without one accountable technical owner.
WHERE	The architecture review board links ADRs, risk records, release gates and exception workflow to the exact artifact and scope.
WHEN	Exercise veto before release or external transfer when a non-negotiable invariant fails; allow exceptions only where policy defines compensating control.
Execution mechanics	An exception states owner, scope, rationale, evidence, compensation, monitoring, expiry and removal plan; it cannot override a prohibited safety action.
Rejected alternative	Consensus-only governance and undocumented verbal waivers were rejected because accountability and expiry vanish.
Failure / degraded mode	A veto freezes the affected alias or capability, not necessarily the entire platform; safe prior paths continue.
Validation evidence	Audit exceptions, expiry, repeated waivers, compensating-control effectiveness and time to remediation.
Revisit trigger	Revisit decision rights when organisational accountability changes, not under delivery pressure.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

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DE-56 Architecture evidence as continuously generated product

Dimension	Principal-architecture position
Decision	Generate diagrams, manifests, ADR links, threat controls, SLO evidence, cost views and traceability matrices from deployed state and registries where possible.
WHY	Static architecture documents drift from implementation and weaken executive, audit and incident decisions.
WHERE	IaC, AWS Config, registries, OpenTelemetry, lineage and workflow events feed an architecture evidence store and controlled report templates.
WHEN	Refresh on deployment, snapshot release, quarterly governance and incident/change review.
Execution mechanics	Evidence distinguishes declared, deployed, tested and observed state; discrepancies become tracked architecture defects.
Rejected alternative	Manual annual evidence collection was rejected because it is expensive, stale and encourages assertion without tests.
Failure / degraded mode	If collection fails, the control is marked unverified; the system does not report compliance from previous evidence beyond its validity.
Validation evidence	Compare generated topology with IaC/runtime inventory, sample traces and verify control-test freshness.
Revisit trigger	Revisit automation coverage as services and regulatory scope evolve.

Decision ownership: the accountable platform owner must be able to identify the released artifact, control point, telemetry, rollback route and human authority for this decision without inferring them from model-generated prose.

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12 Quality-Attribute Scenarios

Quality attributes are expressed as stimulus–environment–response–measure contracts so that platform qualities are testable rather than descriptive.

Scenario	Stimulus and environment	Required response and measure
Scientific replay	Auditor selects a released candidate after source/model retirement	Reconstruct manifest, deterministic transforms, evidence and gates; hashes match or discrepancies are explicitly attributable.
Wrong-direction protection	A high-scoring candidate has conflicting signed mechanism evidence	Direction becomes UNKNOWN; no polarity-dependent promotion; sentinel wrong-direction count remains zero.
Prompt injection	A retrieved paper/tool result contains action instructions and encoded exfiltration	Text remains untrusted data; tool authority unchanged; no external egress; incident telemetry captures attack.
Tenant isolation	Partner-A compound resembles Partner-B hypothesis and shared services are warm	No B data appears in retrieval, cache, feature, trace or training; synthetic cross-tenant precision remains 100%.
Regional failure	Primary Region is lost during snapshot promotion and partner query	Promotion freezes; prior signed alias serves; secondary activates after signature/isolation probes within tier RTO/RPO.
Model regression	Bedrock route changes schema/grounding behaviour	Frozen replay blocks route; current release remains; capability router selects only equivalently evaluated fallback.
Validation integrity	CRO returns result against superseded protocol with incomplete QC	Result quarantined; no label/model update; deviation and reconciliation workflow opens.
Deletion request	Licensed source or partner record must be removed	Lineage-driven deletion reaches chunks, vectors, graph, features, datasets, caches and adapters with completion evidence.
Cost attack	Agent repeatedly expands graph and requests repair	Runtime budgets/cycle detection terminate; partial status disclosed; safety capacity remains reserved.

12.1 Acceptance ownership

Each scenario has one accountable owner, one test harness, one evidence retention policy and one release-blocking threshold. Architecture review rejects qualities stated only as “secure”, “scalable”, “resilient” or “accurate” without a stimulus, measurable response and failure authority.

13 Production Verification Matrix

Control surface	Pre-release verification	Continuous production evidence
Data and licences	Schema/contracts, checksum, admissibility, mapping quality, PIT reconstruction and deletion canary	Release anomaly, lineage completeness, rights-policy decisions and source-impact alerts.
Knowledge graph	Node/edge contract, illegal path, temporal/provenance and tenant predicates	Orphans, path latency, source hydration, poisoning motifs and graph alias integrity.
Features	Golden fixtures, event/availability time, online/offline parity and leakage probes	Drift, null/default rate, freshness, skew and entity-scope violations.
Custom models	Grouped temporal performance, calibration, subgroup/robustness and inference parity	Outcome decay, calibration coverage, feature drift and route-specific defects.
Foundation models	Schema, grounding, contradiction, terminology, injection and cost benchmark	Support precision, veto/repair, token/cost, refusal, latency and route drift.
Agents	Authority, tool parameters, finite termination, state reducers and handoff contracts	Trajectory conformance, tool denials, loops, budget exhaustion and human takeover.
Scientific gates	Sentinel cases, mutation coverage, dependency outage and attempted override	Zero bypass, UNKNOWN preservation, rule-hit distribution and bundle version.
Validation loop	Protocol/result contract, chain of custody, QC, deviation and adjudication	Package reconciliation, label maturity, repeatability and partner SLA.
Security/resilience	Threat campaigns, restore, Region failover, KMS/key and isolation probes	CloudTrail/Security Lake evidence, game-day RTO/RPO, backup/restore and kill-switch tests.

13.1 Evidence retention

Test outputs are content-addressed and linked to the exact candidate snapshot. A dashboard is an operational projection, not evidence authority. Release evidence retains raw test results, harness and fixture versions, environment identity, evaluator artifacts, reviewer/signature records and the policy that interpreted the result.

13.2 Independent verification

Model-risk, quality/regulatory and security verification remain organisationally independent from the implementation owner for high-impact releases. Independence is applied to acceptance judgement and evidence review; it does not require duplicate platform engineering.

14 Failure-Injection and Recovery Programme

14.1 Game-day catalogue

Experiment	Injected fault	Architectural assertion
Snapshot atomicity	Terminate graph/index build after manifest allocation	No partial alias becomes visible; next execution resumes idempotently or rebuilds cleanly.
Bedrock brownout	Latency/5xx on primary author and reviewer routes	Router opens breaker; evaluated alternate or deterministic/human fallback executes without gate weakening.
Neptune/OpenSearch loss	Timeout graph and vector services	Exact structured retrieval and released summaries remain; completeness is downgraded explicitly.
Aurora writer loss	Fail writer during approval/outbox transaction	No split approval/event state; failover preserves transactional invariant and reconciliation.
KMS/access-policy failure	Deny decrypt or inject stale grant	Protected data fails closed; public released artifacts follow approved continuity policy.
Cross-tenant attack	Substitute tenant IDs in tool, cache and trace paths	Every layer denies or isolates; security incident opens on any anomalous access.
Validation tamper	Modify package object/result signature or reorder delivery	Hash/signature fails; artifact quarantines; no training or release update.
Region evacuation	Remove primary Region during active workflows	Promotion freezes, signed serving recovers by tier, callbacks/queues reconcile without duplicate effects.

14.2 Experiment contract

Every experiment declares hypothesis, steady-state scientific metric, blast radius, abort condition, observing owner, expected degraded behaviour and evidence location. Recovery is complete only when decision records, queues, external callbacks, aliases, cost attribution and audit events reconcile—not when infrastructure merely returns healthy.

14.3 Learning loop

A failed assertion creates an architecture defect and may reduce autonomy or resilience tier until re-tested. Incident and game-day findings update sentinel datasets, threat corpora, runbooks, capacity envelopes, ADRs and the decision packets in this document.

15 Principal Architecture Review Checklist

Review axis	Blocking questions
Decision authority	Which fields can models propose, which controls decide, which humans approve, and can any model/tool bypass a veto?
Evidence semantics	Can every claim resolve to admissible source rows/passages with context, date, licence, transform and contradiction status?
Temporal integrity	Are event time, availability time, model cut-off, source release and label maturity enforced across training, replay and serving?
Isolation and rights	Is tenant/purpose/licence policy enforced before retrieval and at every storage, cache, model, trace and export boundary?
Failure semantics	What fails closed, what degrades, what remains read-only, and which prior signed artifact is served?
Evaluation claim	Does the split match intended generalisation; are calibration, subgroup, robustness, leakage and prospective evidence sufficient?
Agent boundedness	Are topology, state reducers, tools, budgets, termination, human takeover and trajectory evaluation explicit?
Operational ownership	Who owns SLO, incident, kill switch, rollback, cost and dependency capacity for each capability?
Regulatory readiness	Are intended use, claims, requirements, verification, electronic approvals, change impact and controlled records linked?
Economics	Can cost be attributed to scientific decisions and validation value without trading away safety or evidence quality?

15.1 Release veto conditions

Release is vetoed for unresolved tenant or licence exposure, untraceable evidence, causal/safety bypass, wrong-direction sentinel failure, invalid point-in-time evaluation, missing rollback, incomplete decision/audit record, unbounded agent authority, unquarantined validation defect or an unowned critical dependency.

15.2 Reconsideration discipline

A rejected pattern may be reconsidered only with new evidence against its original decision criteria. Vendor capability, benchmark improvement or schedule pressure alone does not invalidate an architectural constraint; the new proposal must demonstrate scientific quality, control equivalence, operational recovery, economic value and migration reversibility.